

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 3 Data Science Tools – Python/Libraries

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Notes

We need volunteers to present Python/Timm/PyTorch/IDEs. (send me you name and MST ID)

Will upload the project/homework to Canvas.

TA: Jiaqian Zhu (jzbmn@mst.edu) Office hours: 1:00 – 2:00 pm on Mondays/Wednesdays

PyTorch & IDE: Kaiser, Luke (ltkhg4@mst.edu) Chenevert, William (wucmxv@mst.edu) (Jan. 29th)

Python & Timm & Huggingface: Lewis, James (Jan. 27th)

Data Science Tools – Python/Libraries

- Fundamental Python Libraries for Data Scientists

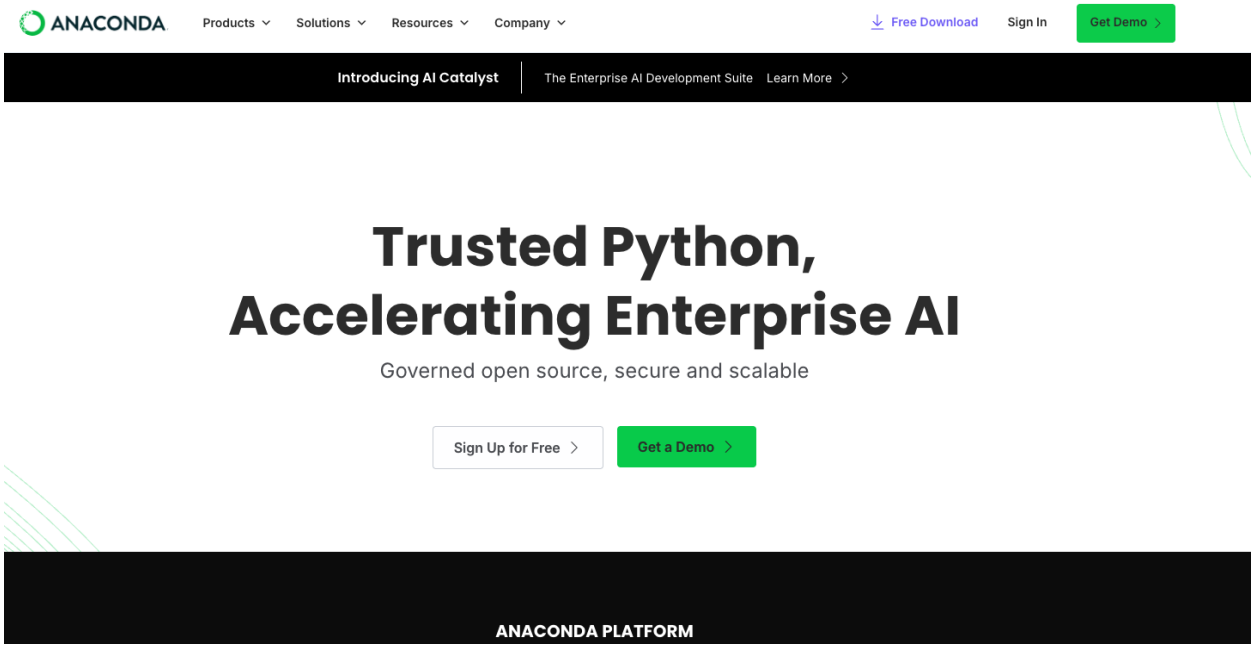
Numeric and scientific computation: Numpy, SciPy, Matplotlib

Machine learning in Python: Scikit-learn and Pytorch

Structured data analysis: Pandas

Data Science Tools – Environment Management

- How to manage python versions and packages?



Data Science Tools – Environment Management

- AI projects may need specific python version or libraries version

  conda create --name <my-env>

```
conda create -n myenv python=3.9
```

Data Science Tools – Python/Libraries

pip install numpy

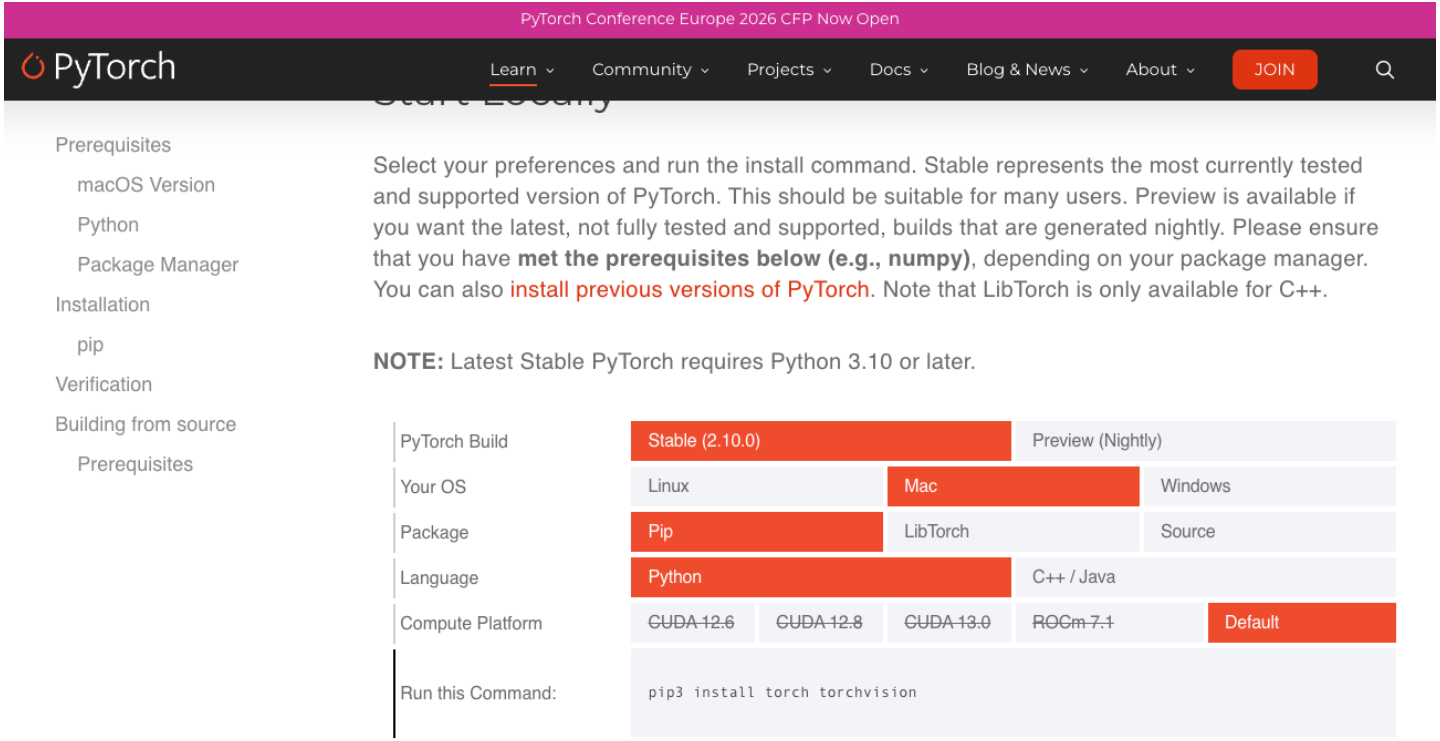
pip install scipy

pip install matplotlib

pip install scikit-learn

pip install pandas

Data Science Tools – Python/Libraries



https://pytorch.org/

Check Libraries in Your Environment

pip list

How to upgrade or downgrade a library?

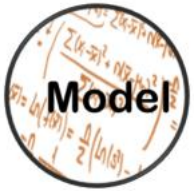
```
(xw312) xiaoweiyu@Xiaoweis-MacBook-Pro IB % pip list
Package                  Version
------------------------ --------------
absl-py                  2.3.1
aiodns                   3.5.0
aiofiles                 24.1.0
aiohappyeyeballs         2.6.1
aiohttp                  3.12.14
aiolimiter               1.2.1
aiosignal                1.4.0
annotated-types          0.7.0
anyio                    4.9.0
anytree                  2.13.0
appnope                  0.1.4
argon2-cffi              25.1.0
argon2-cffi-bindings     21.2.0
arrow                    1.3.0
asttokens                2.4.1
astunparse               1.6.3
async-lru                2.0.5
attrs                    25.3.0
autograd                 1.8.0
azure-common             1.1.28
azure-core               1.35.0
azure-cosmos             4.9.0
azure-identity           1.24.0
azure-search-documents   11.5.3
azure-storage-blob       12.26.0
babel                    2.17.0
```

Data Science Components

Create and use models that learn from data



- Data: anything you can measure or record



- Model: specification of a (mathematical) relationship between different variables



- Evaluation: how well does the model work?

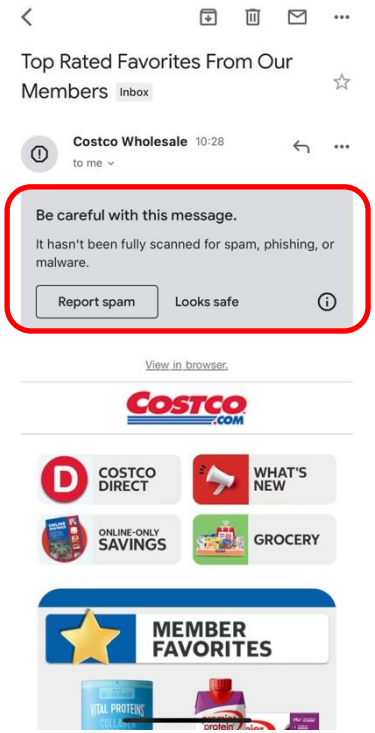
DS problem: Spam Email Classification

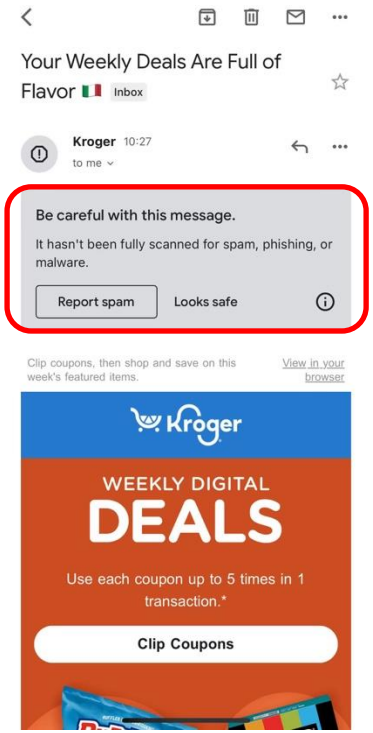
Gmail

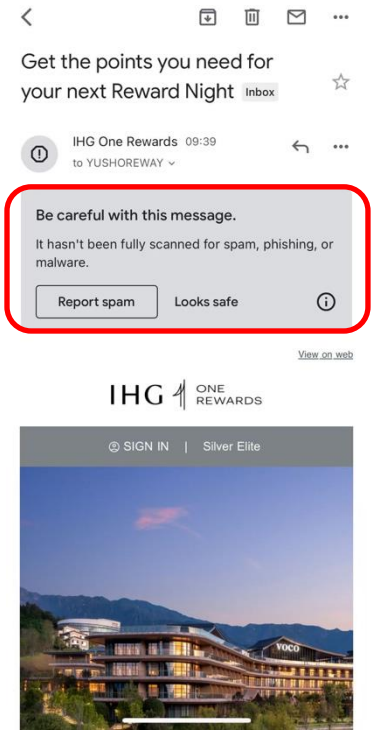
Subscription for websites/applications

Receive advertisements regularly

Collect users' data by feedback







Spam Email Classification

1. Top rated favorites from our members Report Spam Looks Safe

2. Your weekly deals are full of favor Report Spam Looks Safe

3. Get the points you need for your next reward night Report Spam Looks Safe

Gmail active users: 1.8 billion worldwide Roughly 41.9% of the U.S. population uses Gmail.

About 75% Gmail users access email on mobile devices Billions of data are generated daily!

Spam Email Classification – Data Process

Data Samples

Labels

1. Top rated favorites from our members — Report Spam — Looks Safe

2. Your weekly deals are full of favor — Report Spam — Looks Safe

3. Get the points you need for your next reward night — Report Spam — Looks Safe

Use a number, e.g., 1 to represent Spam
User another number, e.g., 0 to represent Safe

(Top rated favorites from our members, Safe)
(Your weekly deals are full of favor, Safe)
(Get the points you need for your next reward night, Spam)

Spam Email Classification – Data Process

Data Samples Labels

(Top rated favorites from our members, Safe)
(Your weekly deals are full of favor, Safe)
(Get the points you need for your next reward night, Spam)



(Top rated favorites from our members, 0)
(Your weekly deals are full of favor, 0)
(Get the points you need for your next reward night, 1)

Use a number, e.g., 1 to represent Spam
User another number, e.g., 0 to represent Safe

Spam Email Classification – Statistics

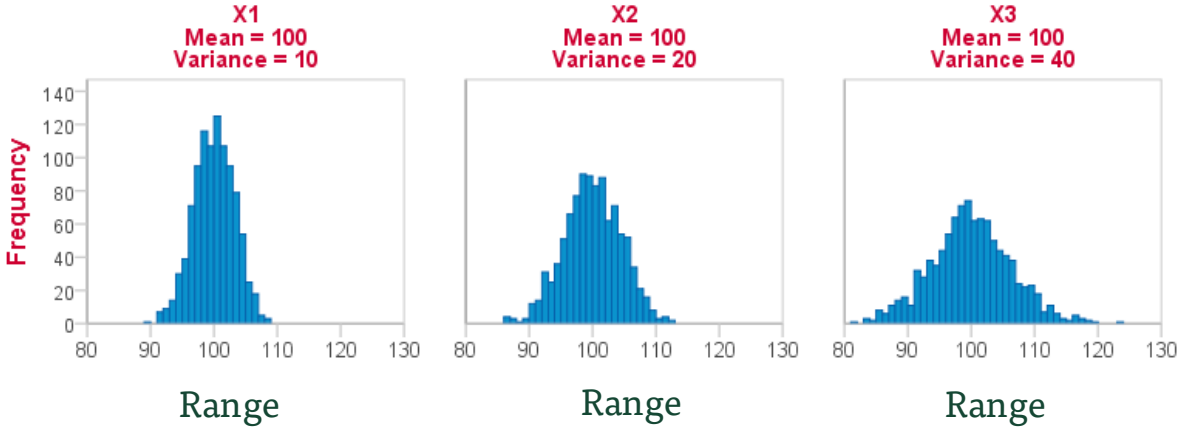
Data Samples	Labels
(Top rated favorites from our members,	0)
(Your weekly deals are full of favor,	0)
(Get the points you need for your next reward night,	1)

Statistics about the data?

Statistics



- Statistics: summarize (and visualize) main characteristics of the data



Mean, Median, STD, Range, Variance, etc..

Spam Email Classification – Statistics

Data Samples Labels

(Top rated favorites from our members, 0)
(Your weekly deals are full of favor, 0)
(Get the points you need for your next reward night, 1)

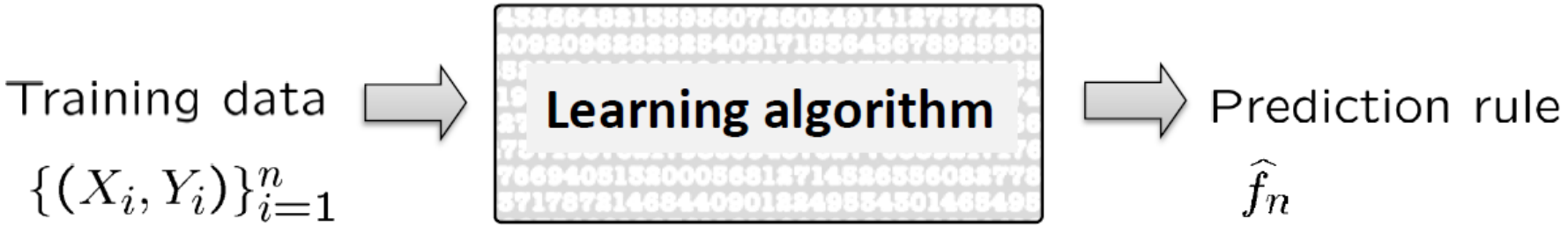
(x, y)

x represent the data samples,
y represents the labels.

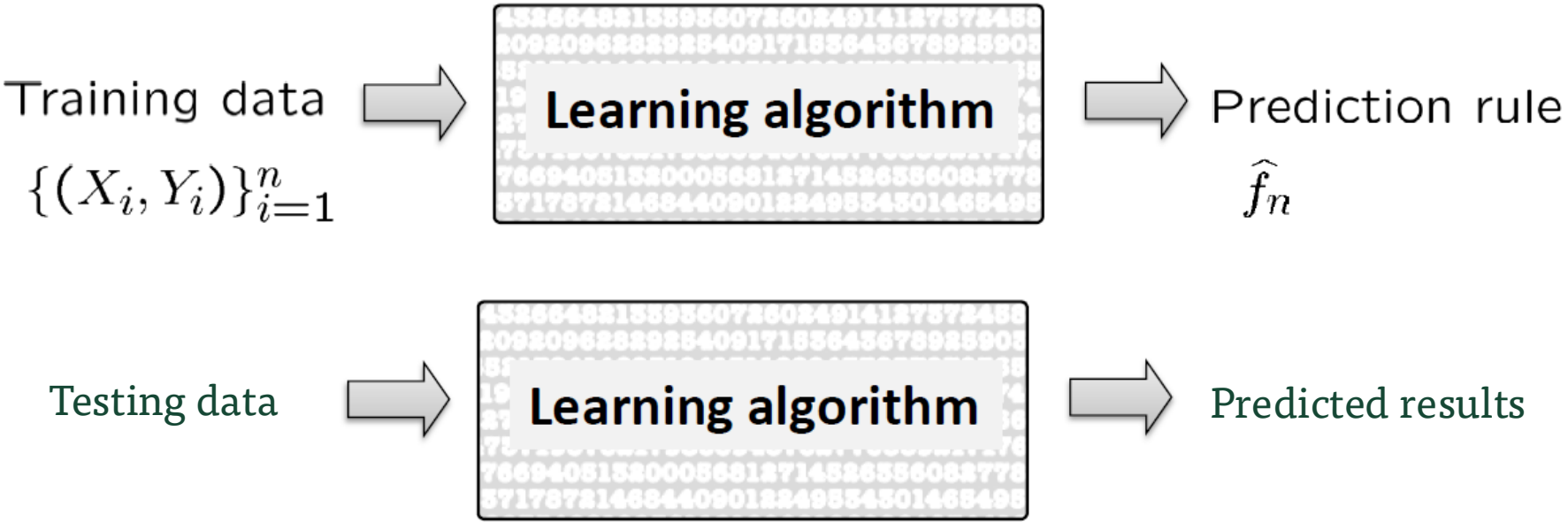
Spam Email Classification – Model

Data Samples	Labels
(Top rated favorites from our members,	0)
(Your weekly deals are full of favor,	0)
(Get the points you need for your next reward night,	1)

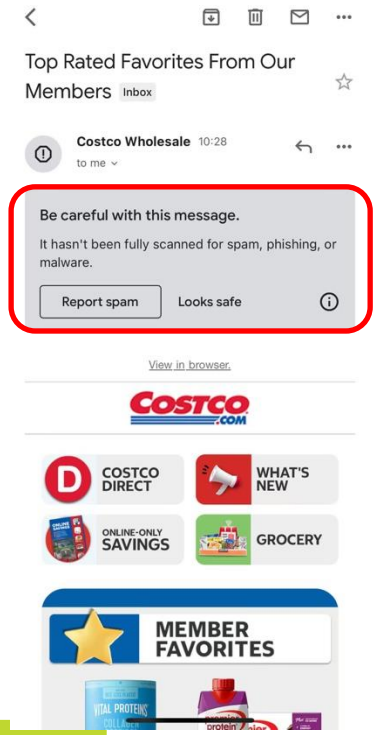
(x, y)

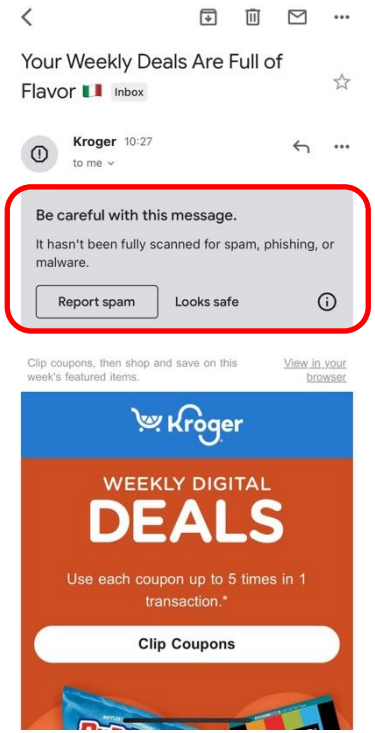


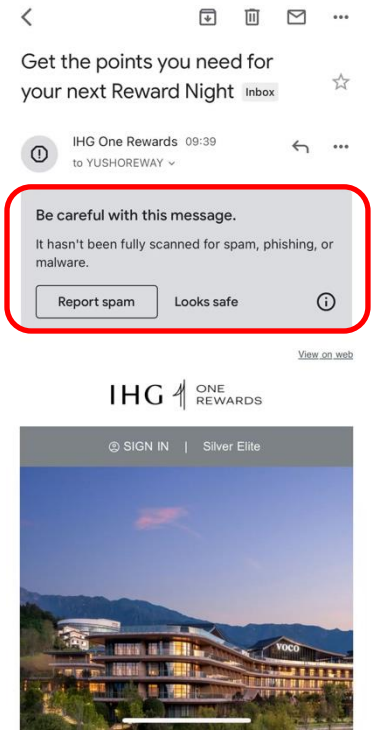
Spam Email Classification – Model



Personalized Spam Email Classification







Personalized Spam Email Classification– Data Process

Data Samples

Labels

1. Top rated favorites from our members — Report Spam — Looks Safe

2. Your weekly deals are full of favor — Report Spam — Looks Safe

3. Get the points you need for your next reward night — Report Spam — Looks Safe

(Top rated favorites from our members, Spam)
(Your weekly deals are full of favor, Spam)
(Get the points you need for your next reward night, Spam)

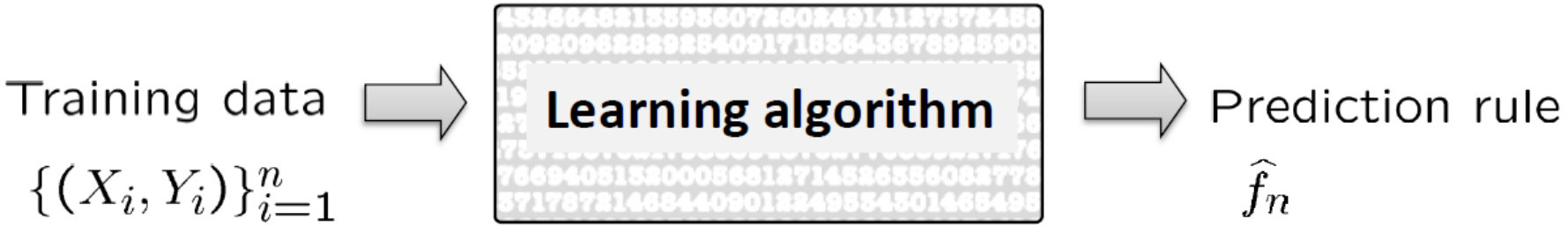
Use a number, e.g., 1 to represent Spam
User another number, e.g., 0 to represent Safe

Personalized Spam Email Classification – Model

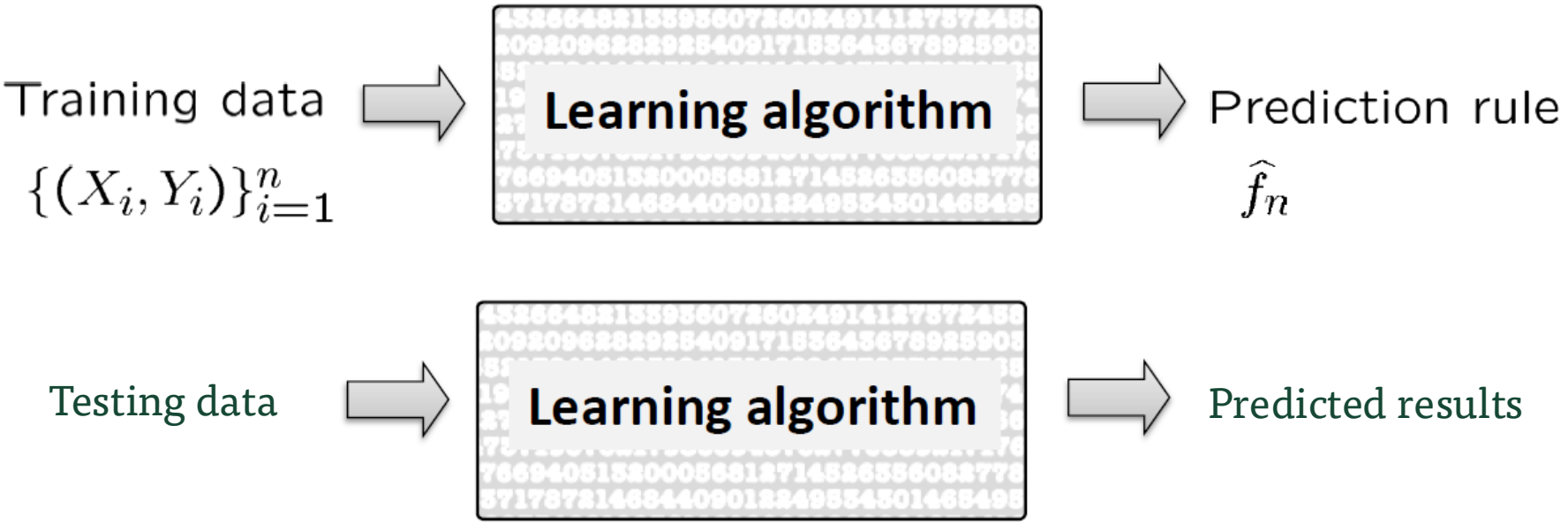
Data Samples Labels

(Top rated favorites from our members, 0)
(Your weekly deals are full of favor, 0)
(Get the points you need for your next reward night, 1)

(x, y)



Personalized Spam Email Classification – Model



Preview

01/29: Data Science Tools – IDE and PyTorch

PyTorch & IDE: Kaiser, Luke (ltkhg4@mst.edu) Chenevert, William (wucmxv@mst.edu) (Jan. 29th)

Homework

Install Anaconda/miniconda.

Create a environment for this course.

conda create --name <my-env>

Install all the necessary libraires under this environment.

Homework

Install Python, Numpy, Matplotlib on your computer.

Relationship between house price and square feet

Sample ID	House Price	Square Feet
1	89, 900	600
2	300, 300	1,576
3	379, 000	3,882
4	1,700,000	11,396

Write python code and use matplotlib to visualize the price and square feet in a plot

Data sources: realtor.com, Homes.com, Zillow.com, Redfin.com

Homework



Sample ID	House Price	Square Feet
1	89, 900	600
2	300, 300	1,576
3	379, 000	3,882
4	1,700,000	11,396

1. Use Python to get statistics of the data

2. Write python code and use matplotlib to visualize the price and square feet in a plot